

Mohit Manna

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EDUCATION

Purdue University

Dec 2027

BS Computer Engineering & Math

Minor: Finance

- **Orgs/Awards:** USACO Gold · 1st/200: Purdue Engineering Symposium · 5x Hackathon Winner

EXPERIENCE

IBM | *Software Engineer Intern*

May – Aug 2026

- Launched a FastAPI LLM triage service for **42** engineers, running **10K** analyses and cutting diagnosis time by **80%**
- Scaled Azure OpenAI across **4** Kubernetes pods with Redis rate limits, sustaining **35** concurrent analyses
- Ingested **32** CI workflows into PostgreSQL at **p95 <5s** with validated asynchronous processing
- Indexed **4,500** CI failures in pgvector with HNSW, retrieving nearest matches in under **5 ms**

Handshake | *AI Engineer*

Jan – Apr 2026

- Trained an RLHF pipeline for code LLMs with Hugging Face on **50K** samples, improving **pass@1** by **18%**
- Built a code-eval harness for **250K** generated programs using async multiprocessing, increasing throughput by **7.4×**
- Optimized AWQ-quantized vLLM inference across **8× A100 GPUs**, cutting **p95** latency by **42%**

Caterpillar | *Machine Learning Intern*

Aug – Dec 2025

- Tuned PyTorch TFT model over **60** Optuna trials, improving supply-chain forecast accuracy by **12%**
- Scaled TorchScript inference across **520** SKUs with Docker & Kubernetes, cutting batch runtime to **2.85 min**
- Automated triggered retraining after a **4-week** shadow deployment, serving forecasts across **20** Angular dashboards

Stealth Startup | *Founding Engineer*

Jun – Aug 2025

- Shipped a React dashboard streaming **10K** real estate data points over WebSockets to **500+** concurrent connections
- Built API infrastructure over **50K** market records, securing **47** Express endpoints with JWT and PostgreSQL RLS

PROJECTS

AegisFeed – High-Performance Market Data Engine in C++ 20

- Benchmarked **23** Nasdaq message types at **8.7M msgs/sec** with zero allocations and bounds-checked parsing
- Stress-tested a bounded **65K**-message reorder buffer under injected packet loss without unbounded memory growth
- Recovered **1.2M** messages under simulated UDP loss at **42 μs p99**, matching clean replay with **0** state mismatches

PulseKV — Low-Latency Distributed Key-Value Engine in C

- Load-tested a concurrent C server across **16** epoll workers at **192K req/sec** on **500** clients at **5.5 ms p99**
- Injected node failures across **256** Raft-backed shards, validating **182K** reads with **0** mismatches

Fintrak – LLM Powered Personal Finance Copilot

- Streamed LLM analysis over **90-day** histories for **200** users tracking **\$180K+** in budgets
- Automated spending alerts above **2× 30-day** baselines, delivering **600+** daily budget nudges

SKILLS

- **Languages:** Python · TypeScript · Rust · C · C++ · Java · SQL · Go
- **Frameworks:** React · Node.js · Express · Pydantic · FastAPI · Flask · Angular · PyTorch · GraphQL
- **Tools:** AWS · Linux · Git · Docker · Kubernetes · Azure · PostgreSQL · Redis · GitHub Actions · OpenTelemetry